

RESEARCH ARTICLE | NOVEMBER 15 2023

Three-phase inverter closed-loop control based on SVPWM drive

Xiangbo Wang 

AIP Conf. Proc. 3017, 040013 (2023)

<https://doi.org/10.1063/5.0173844>



Articles You May Be Interested In

Open loop control of induction motor fed by three-phase inverter based on SPWM and SVPWM techniques

AIP Conf. Proc. (September 2023)

Power factor improvement in simultaneous AC-DC power transmission by using SVPWM compared with SPWM by improving the voltage stability

AIP Conf. Proc. (February 2025)

Comparison between SVPWM two level in internal and external circle of hexagon

AIP Conf. Proc. (January 2017)

Three-phase Inverter Closed-loop Control based on SVPWM Drive

Xiangbo Wang^{1, a)}

¹*Nanjing University of Aeronautics and Astronautics, Nanjing, 210016, China*

^{a)} Corresponding author: xiangbo.wang@nuaa.edu.cn

Abstract. The converter that can convert DC energy (battery, storage battery, etc.) into frequency regulating voltage alternating current or constant frequency regulating voltage alternating current (generally 50Hz sine wave, 220V) is the inverter. In this mobile era, it is applied to all kinds of household appliances and electric equipment. With the great development of power electronics technology and the increasing importance of new energy technology, the application of inverters in professional fields will be more and more in-depth, such as photovoltaic power generation and new energy vehicles and so on. At the same time, in recent years, the relatively novel Space Vector Pulse Width Modulation (SVPWM) technology is widely used in smart grid and power generation technology because of its high voltage utilization and low heat loss. This paper innovatively uses script module programming of ples software to build the SVPWM modulation module which drive the three-phase inverter while realizing the closed-loop control. This research will be beneficial to the application of the new driving mode control inverter in practical production.

INTRODUCTION

As the era gradually entered modernization, the energy structure gradually became sustainable and diversified. However, traditional industrial energy such as oil and coal cannot be used as the core source of energy for mankind permanently due to their limited total amount. At the same time, although the extensive use of fossil energy has promoted the progress of human industrial process, it has a long-term negative impact and harm to the environment. Therefore, in order to solve the above problems, new energy technologies with solar energy and wind energy as the core are gradually becoming large-scale and industrial. In the field of wind power generation and photovoltaic power generation, the inverter is the core device to realize the conversion of electrical signal and the control of electric energy, which has the ability to output high quality voltage waveform.

Xu's paper proposed a control strategy based on SVPWM within the modulation range and improved the overmodulation method, which finally had an effective influence on the control accuracy [1]. Zeng constructed a mathematical model of three-phase inverter with voltage vector tracking current controller based on SVPWM, which has strong control and adjustment ability for photovoltaic system [2]. Legowski proposed a pulse width modulator for a high-speed three-phase inverter, which implements the so-called incremental pulse width modulation algorithm [3].

This paper first introduces the significance of the study of inverter closed-loop control, and then systematically expounds the theoretical basis of SVPWM technology, and introduces the three-phase inverter switch tube state distinction and sector judgment. This paper first introduces the significance of the study of inverter closed-loop control, and then systematically expounds the theoretical basis of SVPWM technology, and introduces the three-phase inverter switch tube state distinction and sector judgment. In the theoretical analysis, the mathematical model is established by means of mathematical equivalent calculation and automatic control principle and applied to simulation analysis. From the open loop control to the final loop control of the output current waveform.

PRINCIPLE DESIGN

Three-phase inverter topology

A simple three-phase inverter topology consists of six Insulated Gate Bipolar Transistor (IGBT) devices. V_{in} is a DC power supply and O is the neutral point of the power supply. L_f and R are the equivalent inductance and resistance values of the output filter inductance while i_a , i_b and i_c indicate the load currents of different branches.

Under pure resistance load, the anti-shunt diode of IGBT is not on-going, and its cycle has 6 different operating modes. There are three switches on at the same time in the six operating modes. One of the two switches of each bridge arm needs to be engaged, and three upper or lower bridge arm switches cannot be engaged at the same time. The topology is shown in figure 1 [4].

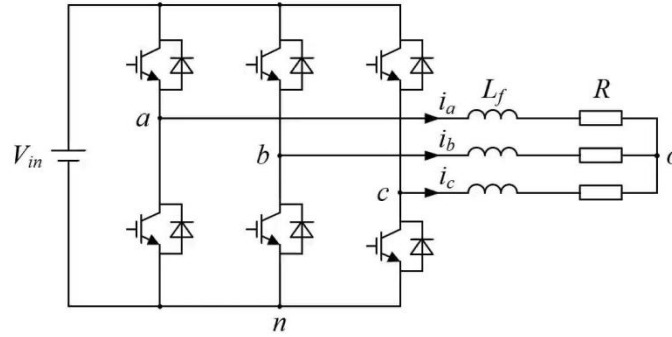


FIGURE 1. Topology of three-phase inverter

In the specific simulation analysis, when selecting the driver module, considering that the switching efficiency of SVPWM modulation strategy is higher and the voltage utilization ratio is 15% higher than that of Sinusoidal Pulse Width Modulation (SPWM) modulation strategy, the SPWM module is used to drive the three-phase inverter.

SVPWM drive technology

The theory and control method of SVPWM has been out for more than 40 years, and in recent years, with the leap forward of power electronics technology, a variety of derivative applications in a mature way to play a more critical role.

From the point of view of device transformation, the essence of SVPWM is that when the inverter arm switch changes, its transmission of electrical signals from different combinations. Then the actual flux produced by different combinations of switching modes is close to the circular flux to the greatest extent, and the real-time switching state of the inverter is determined by their comparison. SVPWM focuses on the idea of considering the operation of the motor and looking at how to make the motor obtain a constant amplitude rotating circular magnetic field. The switch tube conduction status is shown in figure 2 [5].

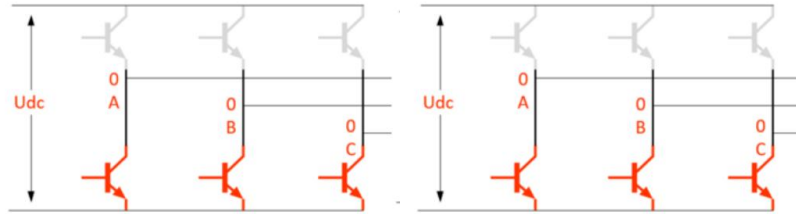


FIGURE 2. Switch tube conduction status

According to the topology structure, the main part of the inverter is composed of six switching tubes. In order to synthesize the space vector of the output voltage with different combination states of the switching tube, the state that the upper bridge arm is on and the lower bridge arm is off can be set as 1, and the opposite is set as 0. There are a total of eight possible combinations of switching tube on-state. It is worth noting that there are two zero vectors and six non-zero vectors. The zero vector has $U_0(000)$ and $U_7(111)$, and the rest are non-zero quantities. Figure 3 shows the position of each switch state vector in space.

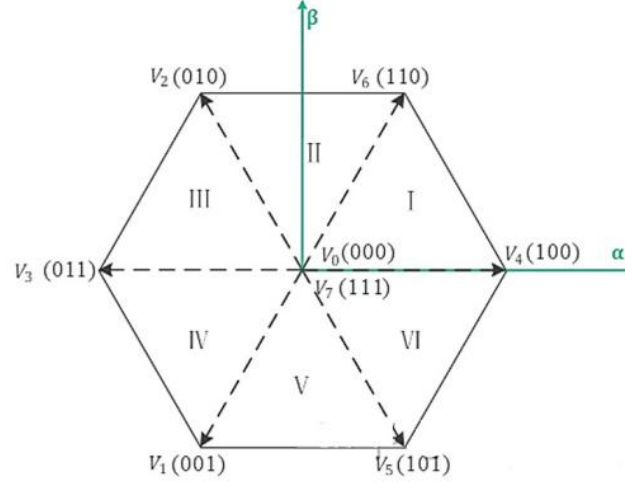


FIGURE 3. Switch tube conduction status

As shown in the figure 3, eight states correspond to eight vectors [6]. These six vectors control power semiconductor -IGBT. When the state of the switch tube changes, such as from off to on, this change will result in heat generation and energy waste. In order to minimize the energy loss, the switch switching of each sector (including the internal sector) needs to ensure that only one action of the bridge arm is changed, so as to minimize the calorific value and achieve higher power density. 100 can be transformed into 110 or 101, which is the difference between positive and negative rotation of the motor. Assuming that the counterclockwise is the forward transmission of the motor, 100 is switched to 110 -- 010 -- 011 -- 001 -- 101 -- 100, forming a cycle. 100-101-001-011-010-110-100, this cycle is the motor to run in the other direction. Any target voltage can be synthesized from these eight vectors.

Open loop design of output current

In order to make the average value of the output phase voltage in the switching period of the three-phase inverter is two specific waveforms after coordinate transformation, as long as the two waveforms are sent to the input module of SVPWM. According to KVL, the above voltage relationship is

$$\begin{aligned} v_{an} &= v_{ao} + v_{on} \\ v_{bn} &= v_{bo} + v_{on} \\ v_{cn} &= v_{co} + v_{on} \end{aligned} \quad (1)$$

According to the theory of coordinate transformation, V_{an} and V_{ao} are exactly equal after transformation. According to the theoretical analysis, the synthesized vector can directly control the values after coordinate transformation of V_{an} , V_{bn} and V_{cn} , which is equal to directly control the values of three-phase coordinate transformation of V_{an} , V_{bn} and V_{cn} . The specific deduction process is as follows:

It can be assumed that the three-phase output current is (f_g is the frequency of the trigonometric function)

$$\begin{aligned} i_a &= i_m \cos(2\pi f_g t) \\ i_b &= i_m \cos(2\pi f_g t - 120^\circ) \\ i_c &= i_m \cos(2\pi f_g t - 240^\circ) \end{aligned} \quad (2)$$

The relation between the inductor current and the inductor voltage is further obtained.

$$\begin{aligned}
v_{ao} &= L_f \frac{di_a}{dt} + Ri_a \\
v_{bo} &= L_f \frac{di_b}{dt} + Ri_b \\
v_{co} &= L_f \frac{di_c}{dt} + Ri_c
\end{aligned} \tag{3}$$

The obtained phase voltage is transformed by Clarke coordinates and continued with mathematical simplification [7].

$$\frac{2}{3}(v_{ao} + v_{bo}e^{j\frac{2\pi}{3}} + v_{co}e^{-j\frac{2\pi}{3}}) = e^{2j\frac{\pi}{3}} i_m (2if_g L_f \pi + r) \tag{4}$$

The concrete function of the theoretical input voltage is obtained by separating the real part and the imaginary part on the axis.

Real part:

$$i_m r \cos(2f_g \pi t) - 2f_g i_m L_f \sin(2f_g \pi t) \tag{5}$$

Imaginary part:

$$2f_g i_m L_f \cos(2f_g \pi t) + i_m r \sin(2f_g \pi t) \tag{6}$$

In the open-loop model, there is no feedback link, but the input directly controls the output. In order to output the ideal load current, just need to calculate and deduce the input waveform directly, which can form a more accurate open-loop control system without interference

Closed-loop design of output current

Next, the transfer function of the controlled system is derived

$$\begin{aligned}
v_{ao} &= L_f \frac{di_a}{dt} + Ri_a \\
v_{bo} &= L_f \frac{di_b}{dt} + Ri_b \\
v_{co} &= L_f \frac{di_c}{dt} + Ri_c
\end{aligned} \tag{7}$$

The three - phase voltage equation is equivalent transformed by Clarke transform

$$\begin{aligned}
v_{ao} &= L_f \frac{di_a}{dt} + Ri_a \\
v_{bo} e^{j\frac{2\pi}{3}} &= L_f \frac{d(i_b e^{j\frac{2\pi}{3}})}{dt} + Ri_b e^{j\frac{2\pi}{3}} \\
v_{co} e^{-j\frac{2\pi}{3}} &= L_f \frac{d(i_c e^{-j\frac{2\pi}{3}})}{dt} + Ri_c e^{-j\frac{2\pi}{3}}
\end{aligned} \tag{8}$$

Add the equations

$$\begin{aligned}
v_{ao} + v_{bo} e^{j\frac{2\pi}{3}} + v_{co} e^{-j\frac{2\pi}{3}} &= [L_f \frac{di_a}{dt} + Ri_a] + [L_f \frac{d(i_b e^{j\frac{2\pi}{3}})}{dt} + Ri_b e^{j\frac{2\pi}{3}}] + [L_f \frac{d(i_c e^{-j\frac{2\pi}{3}})}{dt} + Ri_c e^{-j\frac{2\pi}{3}}] \tag{9} \\
&= L_f \frac{d}{dt}(i_a + i_b e^{j\frac{2\pi}{3}} + i_c e^{-j\frac{2\pi}{3}}) + R(i_a + i_b e^{j\frac{2\pi}{3}} + i_c e^{-j\frac{2\pi}{3}})
\end{aligned}$$

$$v_{\alpha\beta} = L_f \frac{di_{\alpha\beta}}{dt} + Ri_{\alpha\beta} \quad (10)$$

$$v_\alpha + jv_\beta = L_f \frac{d(i_\alpha + ji_\beta)}{dt} + R(i_\alpha + ji_\beta) \quad (11)$$

After the mathematical equivalent transformation of equation (9), the constant amplitude coordinate transformation in Clarke transform is used to decompose the three-phase voltage and current into alpha axis and beta axis, and Equation (10) can be obtained to obtain the corresponding current and voltage values respectively. And then take the Laplace transform of equation (10) to get equation (12) [8].

$$\begin{aligned} v_\alpha &= L_f \frac{di_\alpha}{dt} + Ri_\alpha \rightarrow v_\alpha(s) = sL_f i_\alpha(s) + Ri_\alpha(s) \rightarrow \frac{i_\alpha(s)}{v_\alpha(s)} = \frac{1}{R + sL_f} \\ v_\beta &= L_f \frac{di_\beta}{dt} + Ri_\beta \rightarrow v_\beta(s) = sL_f i_\beta(s) + Ri_\beta(s) \rightarrow \frac{i_\beta(s)}{v_\beta(s)} = \frac{1}{R + sL_f} \end{aligned} \quad (12)$$

This transfer function is the transfer function of the controlled object that implements closed-loop modeling. The block diagram of the overall closed loop is shown in Figure 4 [9].

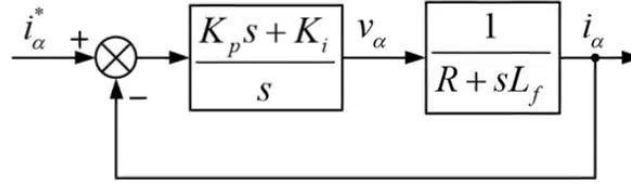


FIGURE 4. Control block diagram of closed loop

The current transfer function of alpha axis and beta axis can be obtained from the above derivation, and then PI control plate can be added according to transfer function. Compare the current reference value with the inductance current sampling value, in which case, the difference is passed into PI control, through the PI link into the SVPWM module and continue to control the three-phase inverter circuit and output the ideal current waveform.

The following is the specific calculation of PI controller parameters K_p and K_i : since the switching frequency f_s is set to 10kHz/s, the crossing frequency f_c can be set to 0.1 times of it, namely 1000Hz/s.

The first equation can be got by $20\lg |G(j\omega)| = 0$. Then make the bode diagram of the controlled object, and then observe the phase-frequency curve to see that it drops from 0° to -90° . After adding PI, it can be seen that the turning frequency point of pi link is $K_i/2\pi K_p$. The phase Angle of PI here is -45° , so it can be pulled down from -90° to -135° at most, that is, the phase Angle margin is sufficient. The PI controller's Bode diagram is shown in figure 5.

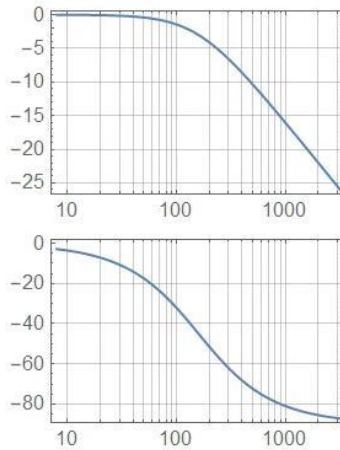


FIGURE 5. PI controller's Bode diagram

The value of K_i and K_p can be calculated using the software mathematical. That is, $K_p = 4.4988$, $K_i = 28266.8$, which and current reference value are brought into the simulation to verify the theoretical analysis

SIMULATION AND ANALYSIS

Introduction to simulation software

Plecs official version is an excellent power electronic system simulator, it has a professional solver and data acquisition system, can quickly simulate a variety of power electronic equipment lines in the simulation analysis. For both industrial developers and academic researchers, this software can accelerate the simulation and analysis of power system, greatly shorten the product development cycle and improve the efficiency of scientific research. In practical application, it is known as "the world's most professional power electronic simulation system".

Mathematica is a kind of scientific counting software, is one of the most widely used mathematical software, is also the most powerful existence in the world general calculation system.

Simulation data setting

In the overall simulation, there are several basic Settings:

- 1.Dc side power supply $V_{in} = 500$ v
- 2.Switching frequency $f_s = 1000$ hz/s
- 3.Input sine wave frequency $f_g = 50$ hz/s
- 4.The maximum step length is set to $1/f_s/100$

Simulation result analysis

The Space Vector PWM module in plecs can directly generate the PWM wave that drives the three-phase inverter, and in the following simulation analysis, we can directly call and drive the three-phase inverter module [10].

In Figure 6, saddle wave is realized by c-script module programming.

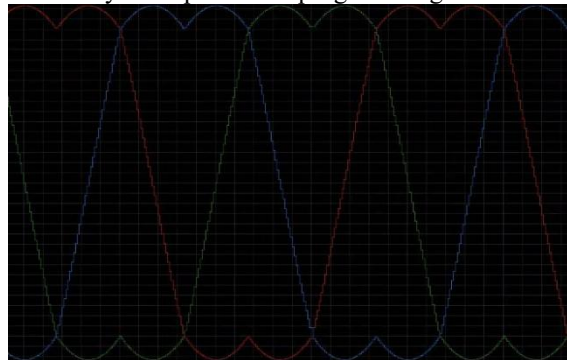


FIGURE 6. Saddle wave shape generated by c-script module programming in plecs

The core idea is to judge the specific sectors of the six sector regions where it falls according to the horizontal and vertical coordinates of the input voltage, so as to realize the ideal output by using adjacent switches to change and change their own states.

In Figure 7, the SVPWM module is called in plecs for simulation.

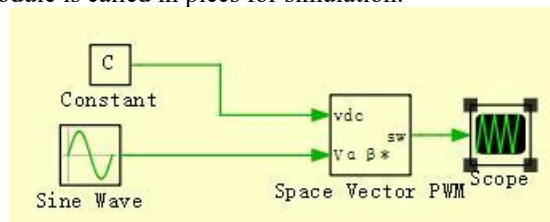


FIGURE 7. Drive signal circuit of three-phase inverter

And the output waveform is observed, that is, the driving waveform of three-phase inverter, as shown in Figure 8.

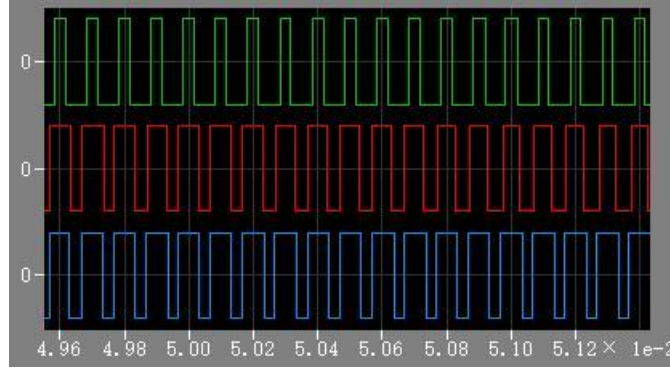


FIGURE 8. ABC driver signal waveform

The following uses simulation to simulate and verify the premise theory of open-loop design. Clarke transforms the mean value of the switching period of three-phase voltage and observes it in the oscilloscope. Figure 9 and figure 10 results verify that if the average switching period value of the output phase voltage of the three-phase inverter is to be two specific waveforms after coordinate transformation, it is only necessary to send these two waveforms to the SVPWM module for input.

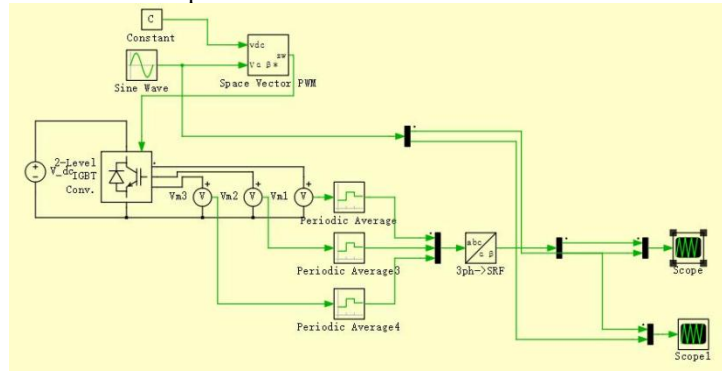


FIGURE 9. Overall simulation of three-phase inverter driven by SVPWM modulation

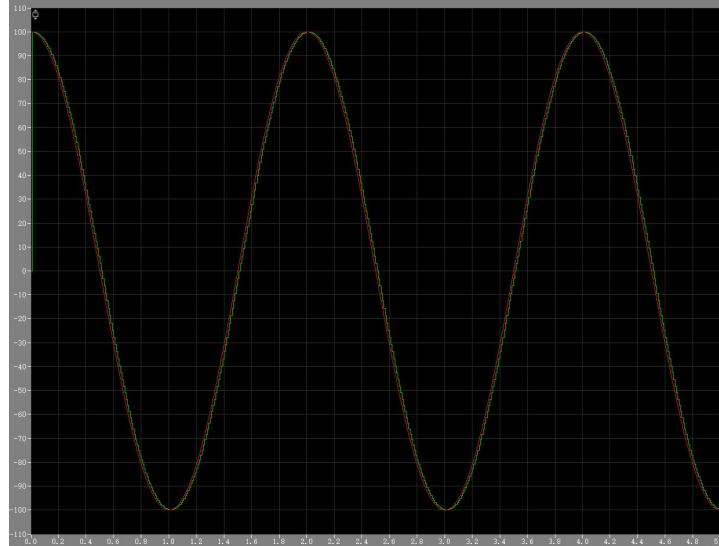


FIGURE 10. The output is compared with the given waveform

At the same time, it is found that the output waveform has a little phase lag compared with the input waveform, which is caused by the average link of the switching period.

The following is the simulation of current open loop control. Input the waveform of the theoretical horizontal and vertical coordinates, and then verify the effectiveness of the open-loop control by tracking the resistance current. Figure 11 and figure 12 shows the output current waveform, and the results are consistent with expectations.

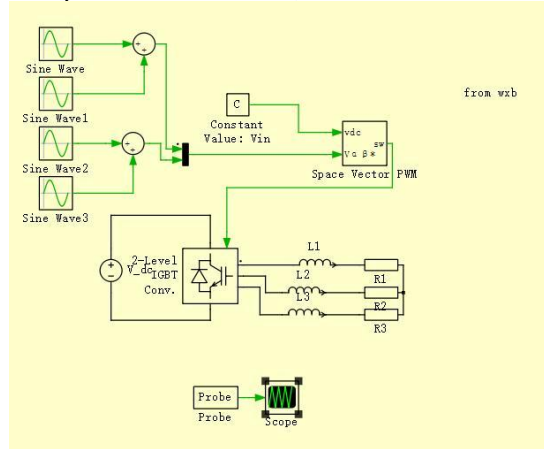


FIGURE 11. Simulation of open loop control of three-phase inverter

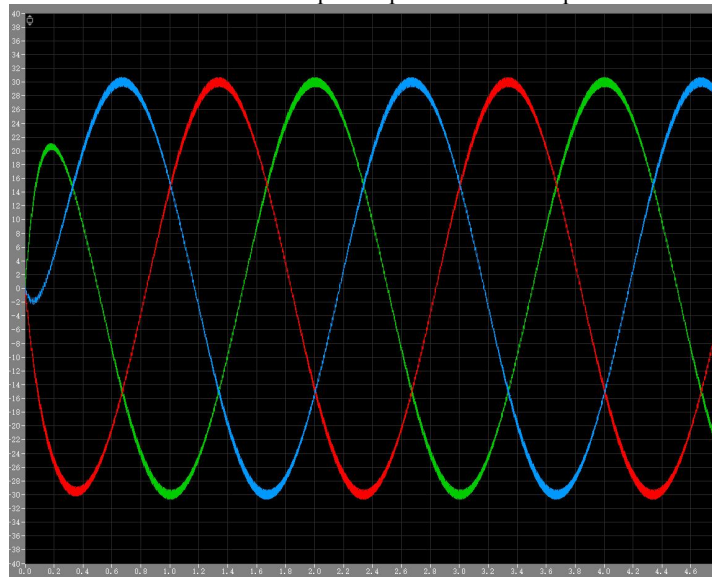


FIGURE 12. Three-phase output current (amplitude 30A)

In this open-loop model simulation, the essence is to advance the want. The current waveform is set, and the expression of the three-phase output voltage is calculated according to the inductance and resistance value. Then coordinate transformation of the three-phase output voltage to obtain the horizontal and vertical coordinates of the two-phase composite vector reference. However, after sending to the SVPWM module, and control the three-phase inverter, and then can produce the expected current amplitude. In practical application, the value of inductance and load resistance is not certain when the inductance value and resistance After the resistance value changes, the input needs to be re-calculated and adjusted accordingly, so the closed-loop model needs to automatically track and Implement control.

Then add a resistance mutation circuit, and observe the three-phase output waveform after the resistance mutation, it can be found that a very large oscillation occurs and amplitude changes obviously after the resistance change as shown in Figure 13.

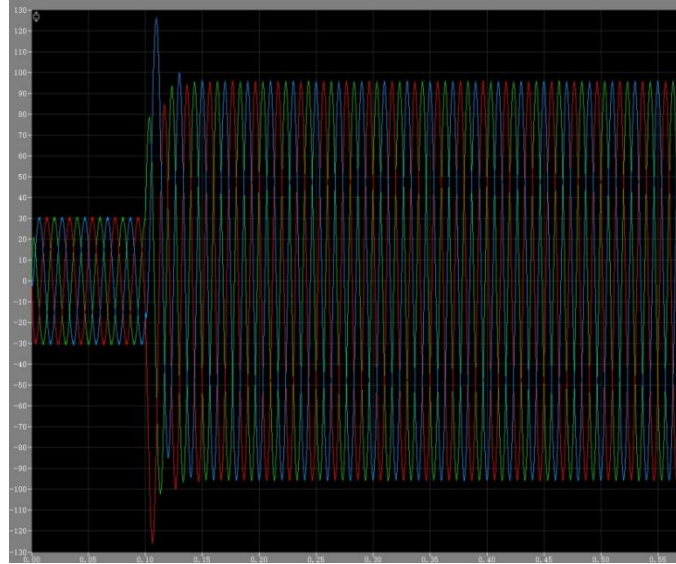


FIGURE 13. Three-phase output waveform after resistance mutation
Then the closed-loop control simulation of the output current was carried out in figure 14 and figure 15.

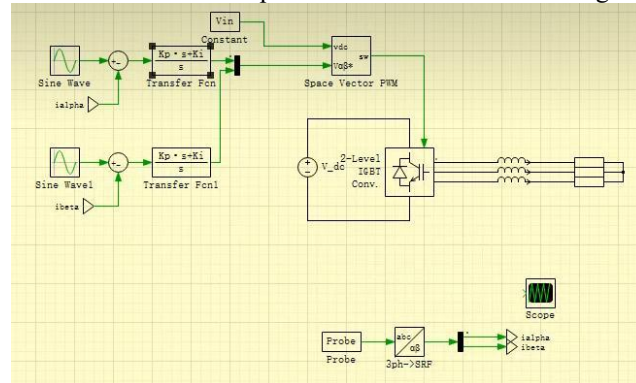


FIGURE 14. Simulation of closed-loop control of output current of three-phase inverter

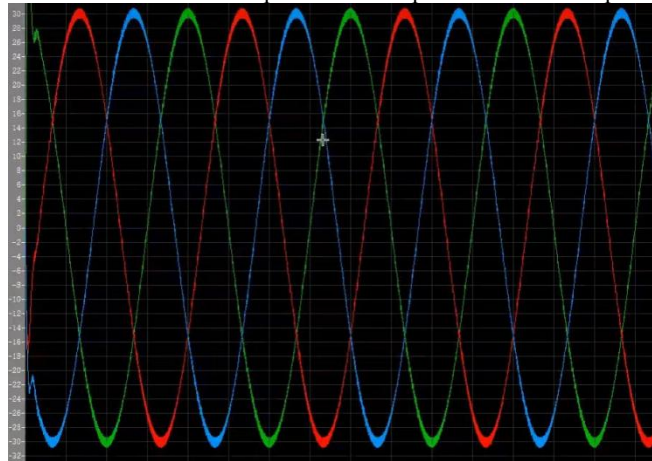


FIGURE 15. Output inductance current waveform

According to the theoretical derivation, a preset PI control link was inserted before the SVPWM module was introduced, and the output current waveform was observed, which was consistent with the expected results.

In order to demonstrate the superiority of closed-loop control, a resistance mutation circuit is connected and observed just like open-loop simulation. Resistance value before and after the output waveform, there is only a little

disturbance. Under the automatic adjustment of PID system, the output current quickly returns to the original steady state as is shown in the figure 16.

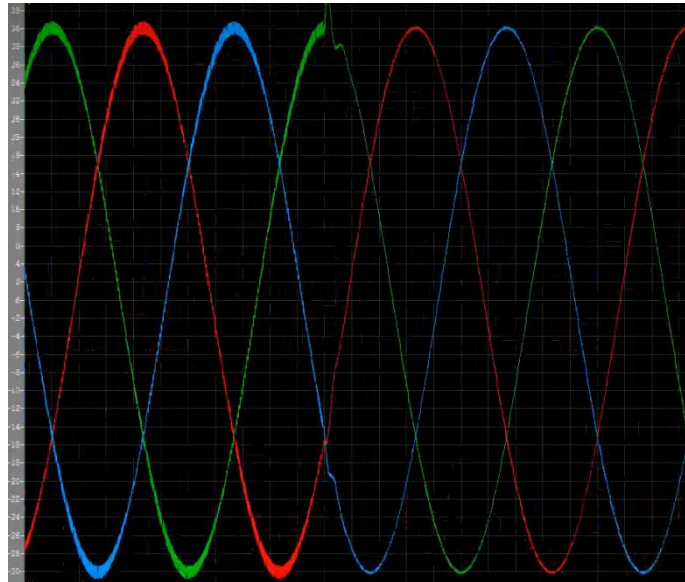


FIGURE 16. The current waveform after the addition of a disturbed circuit

CONCLUSION

It can be clearly found from the simulation experiment that, compared with the open-loop control, the closed-loop control has a strong adjustment ability in the face of disturbance, which can reduce the error and restore the system to the ideal state in a very short time, thus verifying the precision and efficiency of closed-loop PID control. At the same time, the SVPWM modulation technology is systematically used in the theoretical analysis. Compared with SPWM, the voltage utilization rate is increased by 15%, and the switching heat loss is minimal. In conclude, the analysis modeling and simulation methods in this paper are of great significance to the application of inverter in production and life.

REFERENCES

1. Z. Xu, D. Liu, X. Zhao, and J. Ren, "Over-modulation control strategy of SVPWM review," 2016 Chinese Control and Decision Conference (CCDC), May 2016.
2. Z. Zeng, J. Long, and M. Hu, "Research on Three-Phase SVPWM Inverter Based on Double Loop Control of MPPT," Proceedings of the 2015 International Symposium on Energy Science and Chemical Engineering, 2015.
3. S. Legowski and A. M. Trzynadlowski, "Fast Pulse Width Modulator For Three-Phase Inverters," IECON '87: Motor Control and Power Electronics, Oct. 1987.
4. S. Su and W. B. Zhang, "Simulation of SPWM in Three-Phase Inverter Based on LabVIEW and Multisim," [Advanced Materials Research](#), vol. 694–697, pp. 1435–1438, May 2013.
5. E. Zhang, "COMPARATIVE ANALYSIS OF SPWM & SVPWM FED INDUCTION MOTOR DRIVE," International Journal of Advance Engineering and Research Development, vol. 3, no. 05, May 2016.
6. F. Zhao, H. Zhu, and L. Chen, "Research of Grid-connected Inverter Based on SVPWM Control Strategy," Proceedings of the 2016 6th International Conference on Mechatronics, Computer and Education Informationization (MCEI 2016), 2016.
7. Nik Ghazali Nik Daud, Muhazam Mustapha, and Siti Nursyuhada Mahsahirun, "Fast and Efficient Implementation of Forward Clarke Transform's Algorithm Data Flow on FPGA for Green Technology," [International Journal of Digital Content Technology and its Applications](#), vol. 7, no. 5, pp. 400–407, Mar. 2013.

8. D. Wu, S. Kishimoto, and S. Ohnuki, "Optimal Parallel Algorithm of Fast Inverse Laplace Transform for Electromagnetic Analysis," [IEEE Antennas and Wireless Propagation Letters](#), vol. 19, no. 12, pp. 2018–2022, Dec. 2020.
9. A. Upasan and Y. Kumsuwan, "Closed Loop Speed Control of Induction Generator with Scalar-control Inverters," [Energy Procedia](#), vol. 34, pp. 371–381, Jan. 2013.
10. F. Ounis and N. Golea, "PID, 2-DOF PID and Mixed Sensitivity Loop-Shaping Based Robust Voltage Control of Quadratic Buck DC-DC Converter," [Advances in Electrical and Electronic Engineering](#), vol. 14, no. 5, Dec. 2016.